

Guidelines for academic thesis writing at the IMS

Important: **Processing time** of the work is **6 months from registration**

1 Initial pages

1.1 Title page

At the IMS the cover page has to contain:

- name and address of IMS,
- title of thesis,
- name of the author,
- the examiners (2 needed, an examiner can also be a supervisor),
- the supervisor, and
- the date of beginning and end of the thesis.

You may want to use the [LaTeX_Template](#) (by Antje Schweitzer and Nils Reiter) or take a look at its [PDF_Version](#) for reference. Both can also be accessed via [Ilias](#).

1.2 Declaration of authorship

Each thesis has to include a signed declaration of authorship stating location and date. You may want to use the [LaTeX_Template](#) (by Antje Schweitzer and Nils Reiter) or take a look at its [PDF_Version](#) for reference. Both can also be accessed via [Ilias](#).

1.3 Table of contents

The table of contents should contain a complete list of chapters, references, as well as optionally a list of figures, tables, abbreviations and an appendix.

2 Main part

Possible parts can but don't need to include all of the following sections.

2.1 Introduction

For your introduction it can help to try and address as many of the following questions as possible:
(Quelle? Link in Guidelines nicht mehr erreichbar)

- Topic is clearly motivated and introduced
- Goal of research performed in thesis is clear (stating research questions and hypotheses; Can research question be divided into sub-questions? Which ones? Why?)
- Explain what thesis will be about and why reader should care about this topic
- Why is topic interesting? Why is topic relevant for CL?
- Is there actual reason for selecting the topic?
- Explain structure of rest of written document. What is discussed in which chapter/section?

2.2 Background

- All information provided that is required such that another student of same program would be able to understand thesis without additional literature
- Sufficient information on related methods provided such that decision process which led to actual choice of methods is understandable

2.3 Related Work

- Which theories is your work based on?
- What work are you building off of?
- What has already been found so far?
- All work has been described which constitutes are starting point for my own work

2.4 Materials and Methods

- Conceptual and implementation level clearly separated (don't mix snippets with mathematical explanations)
- Implementation details usually not required (if you want to write about them, make section 'Implementation Details')
- Which methods did you use?
- How did you investigate the research question concretely?
- Which data did/do you collect/use/analyse?
- Which hypotheses did you start with?
- Clearly understandable what my own methods are and where I reused what others prepared

2.5 Experiments, Results, Discussion

- Detailed results of experiments/findings
- Experiments are designed to answer the research questions
- Every experiment is clearly explained
- Tables and results are prepared such that reader can easily understand how they help to get an answer to a research question
- Reader would be able to repeat all experiments by themselves with available information
- Reader not left alone with interpreting the results. I talk them through each table and depiction with text
- How do you interpret your findings?
- What does this mean for your research question?
- Could you confirm your hypotheses?
- Could you verify/falsify other work?

2.6 Conclusion

- What was done/discovered?
- What are implications of this work?
- I answer the research questions posed in the introduction
- I summarize main take home messages
- Briefly summarize main results once more:
 - What did I find?
 - What was new, what was unexpected?
 - Where the methods appropriate?
 - What problems occurred?
 - Motivate other research and give an outlook: How could research question be investigated further in the future? How could your results be applied/verified or refined?
 - What else might be relevant in this research area?
 - What future benefit could come from your work?
 - What (if anything) still remains to be done?
 - What are open research questions?
 - **All questions mentioned in introduction should be addressed again in conclusion**

3 Final part

3.1 Bibliography

- Every entry is complete: all information available and easy for reader to understand to which source an entry corresponds
- ArXiv papers (and similar) only cited if there is no published version of paper available (for example in ACL anthology)
- All entries are consistent: author's first names are always (or never) abbreviated; abbreviation always have a '.' behind entry (or never), conferences are called what they are called in official publication, journals have numbers and volumes; page numbers included when available
- **Correct and consistent style/format:**
 - Citations should be done using BibTeX or some reference manager for Word or Open Office/Libre Office (e.g.: Citavi, EndNote, Zotero)
 - All relevant fields in bibliography database are filled out (all authors, journal/conference, year, page numbers,...) and have consistent capitalization
 - In-text citations are all in consistent format
- **Correct and complete citations:**
 - All references in reference section should be cited somewhere in thesis
 - Content of source actually matches to how you want to use the citation
 - Any figures taken from other sources are correctly cited

3.2 Appendix (optional)

The appendix can but does not need to contain:

- code or excerpts from code,
- transcripts of interviews,
- (annotation) guidelines
- list of figures/abbreviations/...
- etc.